

☐ Movie Recommendation System

This is a simple **AI/ML-based Movie Recommendation System** built using Python.

The project recommends movies that are similar to a movie selected by the user.

The system uses **TF-IDF Vectorization** and **Cosine Similarity** to compare movie genres and descriptions and generate recommendations.

☐ Features

- ☐ Select a movie from the available movie list
 - ☐ Uses machine learning techniques for recommendations
 - ☐ Analyzes movie genre and description
 - ☐ Converts text into numerical features using **TF-IDF**
 - ☐ Finds similar movies using **Cosine Similarity**
 - ☐ Shows the top 5 recommended movies
 - ☐ Displays a similarity percentage for each recommendation
 - ☐ Simple Python implementation that can run directly in Google Colab
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☐ How the System Works

1. Movie Data

The project contains movie information such as:

- Movie title
- Genre
- Description

The genre and description are combined to create the movie's feature information.

2. TF-IDF Vectorization

The movie features are converted from text into numerical values using:

TF-IDF (Term Frequency--Inverse Document Frequency)

This allows the computer to understand which words are important when comparing movies.

3. Cosine Similarity

The system compares the numerical movie vectors using **Cosine Similarity**.

A higher similarity value means that two movies have more similar features.

4. Recommendation

After the user enters a movie name:

```
User selects a movie
      ↓
Movie features are selected
      ↓
TF-IDF converts text into vectors
      ↓
Cosine Similarity compares movies
      ↓
Movies are sorted by similarity
      ↓
Top 5 movies are recommended
```

☐ AI / ML Used

The main machine learning concepts used in this project are:

- **Content-Based Filtering**
- **TF-IDF Vectorization**
- **Cosine Similarity**
- **Natural Language Processing (NLP)**

The system follows a **content-based recommendation approach**, because recommendations are generated from the characteristics of the selected movie.

☐ Project Structure

```
Movie-Recommendation-System/
|
├─ movie_recommendation.py
└─ README.md
```

For the Google Colab version, the complete Python code can simply be placed inside a notebook cell.

▶ How to Run Google Colab

1. Open Google Colab.
2. Create a new notebook.
3. Run the required libraries:

```
!pip install pandas scikit-learn -q
```

4. Paste the Movie Recommendation System code into the next cell.
5. Run the cell.
6. Enter a movie name when prompted.

Example:

```
Enter a movie you like: Inception
```

□ Sample Output

❑ MOVIE RECOMMENDATION SYSTEM

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Available Movies:

- Inception
- Interstellar
- The Dark Knight
- The Matrix
- Iron Man
- Avengers Endgame
- Avatar
- Jurassic Park
- Titanic
- The Shawshank Redemption

Type 'exit' to close the program.

Enter a movie you like: Inception

❑ Recommended Movies

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1. The Matrix (similarity %)
2. Interstellar (similarity %)
3. Avatar (similarity %)
4. Iron Man (similarity %)
5. Jurassic Park (similarity %)

The exact similarity percentages can change depending on the movie data and feature descriptions.

❑ Technologies Used

Technology Purpose

Python Main programming language
Pandas Managing movie data
Scikit-learn Machine learning calculations
TF-IDF Text feature extraction
Cosine Similarity Movie similarity calculation
Google Colab Running and testing the project

❑ Example

If the user enters:

Inception

the system looks at the features of **Inception** and compares them with the other movies.

Because movies such as *The Matrix* and *Interstellar* contain related **Sci-Fi / Thriller / Adventure** features, they can receive higher similarity scores.

☐ Future Improvements

The project can be expanded by adding:

- ☐ User accounts and movie ratings
 - ☐ Personalized recommendations based on rating history
 - ☐ Collaborative Filtering
 - ☐ A larger real-world movie dataset
 - ☐ Movie posters and details
 - ☐ Movie search
 - ☐ Web interface using Flask or Streamlit
 - ☐ Hybrid recommendation system combining content-based and collaborative filtering
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☐ Project Objective

The main objective of this project is to demonstrate how **AI/ML techniques can be used to build a recommendation system**.

Instead of manually selecting movies, the system analyzes movie information and automatically finds movies with similar characteristics.

☐ ☐ Conclusion

The Movie Recommendation System demonstrates a practical application of **machine learning and NLP**.

By using **TF-IDF** and **Cosine Similarity**, the system converts movie descriptions into machine-readable features and recommends movies based on their similarity.

